

Queens of the Castle - Game Description

Students try to place as many queens as possible on a chessboard without any of them attacking each other. Once they have found a solution in their workbooks, they will test it on a human sized chessboard!

Resource List

- Printed Student Workbooks (1 per group)
- Counters (8 per group) [optional]
- Pens (around 2 per group is preferable)

- *Printed Tutor Instructions & Answers (approx 1 per 3 tutors)*
- *Student extension workbooks (1 per group)*
- *Tutor Variation Record (1 per big grid)*

- Tape for making floor grids
- Cones acting as the 'Queens' (8 per big grid) [optional, can use tutors and group members to fill the board]
- Computerless rewards (Freddos/Mentos)

Setup - Giant grid

If space allows, we want to construct a human-sized 8x8 chessboard. Here's some ideas on how to construct it, but do whatever makes sense for your space. 1 grid should be enough but if you have the space and the tutors it can be fun to make more.

Get tutors to help set up the grid.

Option 1: Paper squares:

Resources: 32 pieces of A4 paper (per chessboard)
 Blue tack
 Check the ground of the area you are using to see if blue tack will stick to it

Steps:

1. Cut down the long side of the piece of paper, so it is as long as the short edge (creating a square).
2. Tile the pieces of paper in a checkerboard pattern by leaving one empty square between each piece of paper. (each row will have 4 paper squares, don't forget that empty square on the end!)
3. Blue tack all the papers to the ground
4. Use the offcuts or more paper to label the axes

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Option 2: Tape:

Resources: Approx. 36 meters of masking tape (for a 2x2 meter grid)
Check the ground of the area you are using to see if tape will stick to it
Paper may help you set the spacing.

Steps:

Use a ruler, measuring tape or pieces of paper to get consistent spacing. The final grid will be about 2x2 meters.

1. Tape out an 8x8 grid on the ground by taping 9 vertical lines spaced approx. 25-30cm apart horizontally.
2. Repeat vertically.
3. Label the axes - you could do this by bluetacking paper, or using marker on the tape etc.

Bonus: use an A4 piece of paper cut down to a square as colouring for the alternating squares. Arrange the pieces of paper before taping the grid out.

Option 3: Jumbo Chalk:

Resources: Jumbo chalk (or one of those chalk line drawers builders use)
Only do this on a surface you are allowed to chalk on (probably outside)

Steps:

Use a ruler, measuring tape or pieces of paper to get consistent spacing. The final grid will be about 2x2 meters.

1. Chalk out an 8x8 square grid on the ground by taping 9 vertical lines spaced approx. 25-30cm apart horizontally.
2. Repeat vertically.
3. Label the axes

Tutors

At the start of the activity try to have a tutor assigned to handing out each resource for each team (e.g. 1 on the worksheets, 1 on counters, 1 on pens) to help streamline the handout process.

If possible aim to have ~6 tutors per big grid, at least 1 tutor can focus on checking the teams previous solutions, at least 1 can focus on distributing the extension worksheets and rewards, but all tutors outside should help model the group's 8-queen solution, encouraging students to use the grid coordinates to direct them.

The remaining tutors should be floating around helping groups if there are any issues or confusion.

[may want to assign a lead of each grid - they will check/record the solution]

[Have tutors help set up the giant grid]

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Students

Students should complete the worksheets in one location, before moving to get their 8 queens solution checked on the giant grid. Preferably in groups of 4.

Running the Game

Groups will need:

- 1x handout
- 2x pen
- 8x counters [optional]

Students can use counters to model possible solutions, and then write down the co-ordinates when they are happy with their answer. →

Once they think they have an answer, they should take the co-ordinates to the big grid, where tutors can test it. →

If their solution works, they receive a reward and collect the extension.

When the students finish their handout they will take it to the big grid and

- Get their earlier solutions checked by a tutor,
- Model their 8 queens solution on the big grid,
- Receive a reward,
- Receive the extension handout.

Finishing the Game

To wrap up this game, draw everyone's attention back to the front of the room.

[There will be slides - reveal all the various solutions they managed to find. Reflect on any patterns they may have noticed (ask the audience for their observations) - relate this to candy catcher? E.g. searching co-ords]